

revolutionised lives all over the world, but, it is true, that it also has some fears associated with it, which will need to be allayed.

7.4 Wireless sensor network

A wireless sensor network (WSN) is a set of autonomous machines that use in-built sensors to cooperate, and together monitor physical phenomena such as sound, temperature, pressure, etc. Like all networks, a WSN also consists of individual components called nodes. In a WSN, each node contains the required sensor, a microcontroller, a battery and a communications device such as a radio transceiver.

Wireless sensor networks were initially built as a military tool and used as a surveillance mechanism

Apart from the nodes, there needs to be a control unit in the WSN. This unit is in charge of processing all the data that is gathered by the various sensors which are in operation. It is also usually programmed to analyse the data gathered and thus reconstruct the required phenomenon.

Some of the characteristics of the WSN make it unique. Nodes in the WSN are built for specific conditions and thus can withstand extremes in environmental conditions. In a WSN, the failure of one node does not lead to the failure of the system. The rest of the nodes will continue to function and will all do their job. Another unique aspect is the dynamic network topology. Being wireless, the nodes can be shifted anywhere and they can also keep exchanging roles without a problem. This is an advantage when something is being monitored that changes rapidly. The biggest advantage is that the operation of a WSN does not need to be constantly monitored, and this makes it suitable for use in conditions where human interference is not possible.

However, there are also some problems with a WSN system. The biggest is that the power source that a node uses is a battery. This makes the node functionality limited to the battery life. There is also higher risk of communication failures because of the environment, and these are also harder to repair.

The battery constraints make it necessary that the programs in use in the system are all fast, efficient and focussed. The